1) $h(n) = (1 \quad 2 \quad 1)$

$H(z) = 1 + 2z^{-1} + z^{-2}$ $\qquad z = e^{j2\pi f T}$ $\qquad T = \frac{1}{r} = 1$

$H(f) = 1 + 2e^{-j2\pi f} + e^{-j4\pi f}$

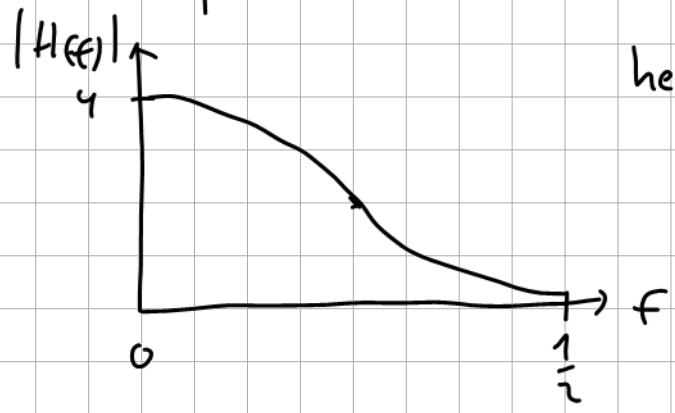
$= e^{-j2\pi f}\left(e^{j2\pi f} + 1 + e^{-j2\pi f}\right)$

$|H(f)| = \left|e^{j2\pi f} + 1 + e^{-j2\pi f}\right|$

$= \left|\cos(2\pi f) + j\sin(2\pi f) + 2 + \cos(2\pi f) - j\sin(2\pi f)\right|$

$= \left|2 + 2\cos(2\pi f)\right|$ $\qquad 0 \le f \le \frac{r}{2}$ Sampling-theorem

here : $0 \le f \le \frac{1}{2}$



$\Rightarrow$ $h(n)$ is a lowpass.

2) Converting a Lowpass to an Highpass:

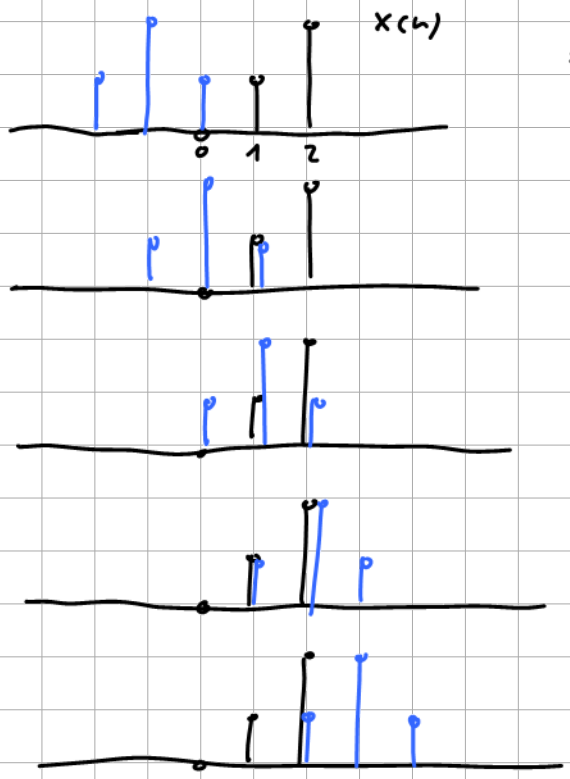
$$h_{HP}(n) = h_{LP}(n) \cdot (-1)^n$$

$$h_{LP}(n) = 1 \quad 2 \quad 1$$

$$h_{HP}(n) = 1 \quad -2 \quad 1$$

3) $x(n) = (0 \quad 1 \quad 2)$ $\qquad$ $h(n) = (1 \quad 2 \quad 1)$

graphical method:



$\Rightarrow \quad y(0) = 1 \cdot 0 = 0$

$\Rightarrow \quad y(1) = 1 \cdot 1 + 2 \cdot 0 = 1$

$\Rightarrow \quad y(2) = 1 \cdot 2 + 2 \cdot 1 = 4$

$\Rightarrow \quad y(3) = 1 \cdot 1 + 2 \cdot 2 = 5$

$\Rightarrow \quad y(4) = 1 \cdot 2 = 2$

$y(n) = 0$ everwhere else !

4)

$$h_{LP}(n) = \frac{\sin\left(2\pi f_c \frac{n}{r}\right)}{2\pi f_c \frac{n}{r}} \cdot \frac{1}{2}\left(1 + \cos\left(2\pi \frac{n}{N}\right)\right)$$

with: $-2 \leq n \leq 2$

$$\frac{f_c}{r} = \frac{1}{4}$$

$$h_{LP}(n) = \frac{\sin\left(\frac{\pi}{2} n\right)}{\frac{\pi}{2} n} \frac{1}{2}\left(1 + \cos\left(2\pi \frac{n}{5}\right)\right)$$

$$h_{LP}(-2) = h_{LP}(2) = 0$$

$$h_{LP}(-1) = h_{LP}(1) = 0{,}417$$

$$h_{LP}(0) = 1$$

5)

$$y(n) = x(n) * h(n)$$

$$= \sum_{k=0}^{K-1} x(n-k) \cdot h(k) \qquad K = 501$$

$\Rightarrow$ K Multiplications

$K-1$ Additions

6) minimize $\sum_n \left(a \cdot y(n-T) - z(n)\right)^2 = f(a)$

first derivative: $\sum_n 2\left(a\, y(n-T) - z(n)\right) \cdot y(n-T) = \frac{df(a)}{da}$

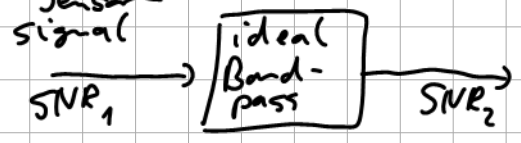
setting to 0: $\sum_n 2\left(a\, y(n-T) - z(n)\right) \cdot y(n-T) = 0$

$$\sum_n a\, y^2(n-T) - z(n)\, y(n-T) = 0$$

$$\sum_n a\, y^2(n-T) = \sum_n z(n)\, y(n-T)$$

$$a = \frac{\sum_n z(n)\, y(n-T)}{\sum_n y^2(n-T)}$$

7)



$$SNR_1 = 10 \log_{10} \frac{P_x}{P_{R_1}}$$

$$SNR_2 = 10 \log_{10} \frac{P_x}{P_{R_2}}$$

increase in SNR: $SNR_2 - SNR_1 = 10 \left[\log_{10} \frac{P_x}{P_{R_2}} - \log_{10} \frac{P_x}{P_{R_1}} \right]$

$$= 10 \cdot \left[\log_{10} P_x - \log_{10} P_{R_2} - \log_{10} P_x + \log_{10} P_{R_1} \right]$$

$$= 10 \cdot \log_{10} \frac{P_{R_1}}{P_{R_2}}$$

power of a signal is the integral over the power density function:

$$P_{R_1} = \int_{0\,Hz}^{24000\,Hz} |R(f)|^2 \, df = \int_{0\,Hz}^{24000\,Hz} c \cdot df$$

($|R(f)|^2$ is assumed to be constant)

$$= c \cdot 24000$$

$$P_{R_2} = \int_{100\,Hz}^{7000\,Hz} |R(f)|^2 \, df = 6900 \cdot c$$

SNR gain (increase) $= 10 \log_{10} \frac{24000c}{6900c} = 10 \log_{10} \frac{240}{69} = 5{,}4\ dB$

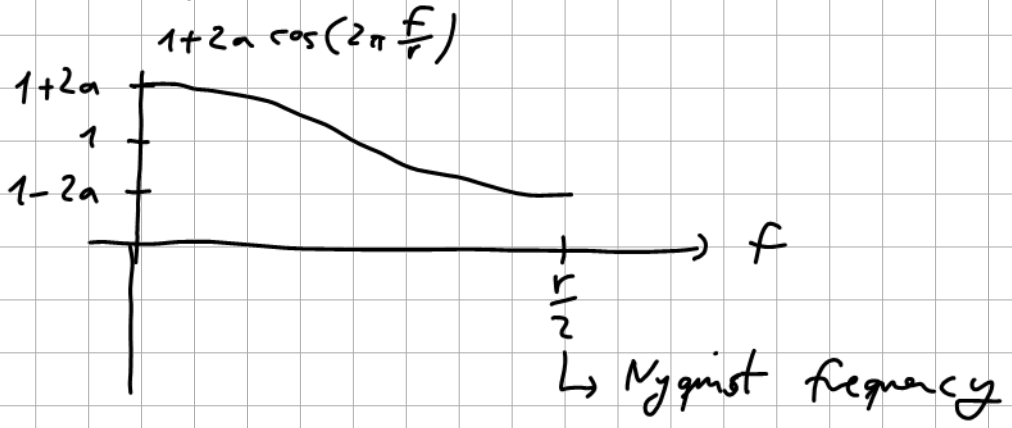
8) $h(n) = [a \quad 1 \quad a]$

$$H(f) = a + e^{-j2\pi \frac{f}{r}} + a e^{-j4\pi \frac{f}{r}}$$

$$= e^{-j2\pi \frac{f}{r}} \left(a e^{+j2\pi \frac{f}{r}} + 1 + a e^{-j2\pi \frac{f}{r}} \right)$$

$$|H(f)| = \left| a e^{j2\pi f/r} + 1 + a e^{-j2\pi \frac{f}{r}} \right|$$

$$= \left| 1 + 2a \cos(2\pi f/r) \right|$$



Condition: $1 - 2a > 0 \Rightarrow 1 > 2a \Rightarrow \frac{1}{2} > a$

9) $h(n) = [-1, 1]$

- A FIR filter is always stable.
- $H(f) = -1 + e^{-j2\pi fT} = -1 + e^{-j2\pi f/r}$

$Re\{H(f)\} = -1 + \cos(-2\pi f/r) \leq 0$ for all $f \in \mathbb{R}$

10) $h(n) = [1, 4, 6, 4, 1]$ $\qquad r = 48\ kHz$

$= h_{LP}(n)$ $\qquad f_m = 12\ kHz$

$$h_{BP}(n) = h_{LP}(n) \cdot \cos(2\pi f_m \cdot n / r)$$

$$= [1, 4, 6, 4, 1] \cdot \cos\left(2\pi n \cdot \frac{12\ kHz}{48\ kHz}\right)$$

$$= [1, 4, 6, 4, 1] \cdot \cos\left(\frac{\pi}{2} \cdot n\right)$$

n	$h_{LP}(n)$	$\cos\left(\frac{\pi}{2} \cdot n\right)$	$h_{BP}(n)$
0	1	$\cos\left(\frac{\pi}{2} \cdot 0\right) = 1$	1
1	4	$\cos\left(\frac{\pi}{2} \cdot 1\right) = 0$	0
2	6	$\cos\left(\frac{\pi}{2} \cdot 2\right) = -1$	-6
3	4	$\cos\left(\frac{\pi}{2} \cdot 3\right) = 0$	0
4	1	$\cos\left(\frac{\pi}{2} \cdot 4\right) = 1$	1

~~11)~~

11)

$$y(n) = a\,y(n-1) + (1-a)\,x(n)$$

z-Trafo

$$Y(z) = a z^{-1} Y(z) + (1-a)\,X(z)$$

$$Y(z)\,(1 - a z^{-1}) = (1-a)\,X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1-a}{1-az^{-1}} = \frac{(1-a)\,z}{z-a}$$

$\Rightarrow$ pole at $z = a$

$\Rightarrow$ stable for $|a| < 1$

$\Rightarrow$ instable for all other a.

14 ~~8~~)

The mean value of a signal corresponds to the DC part of a signal.

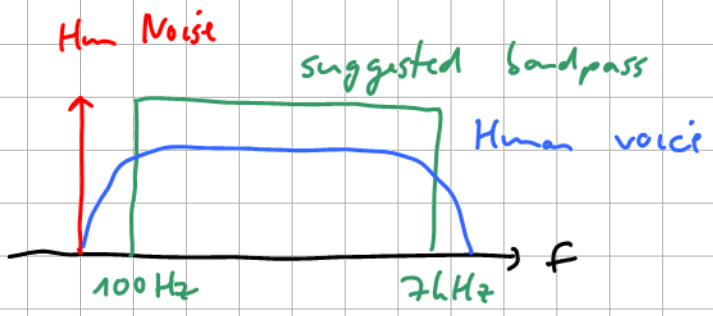
This DC part corresponds to the frequency $f = 0\,Hz$.

After a bandpass, no DC-component remains.

$\Rightarrow$ The signal has a zero mean after the bandpass.

Wideband human speech (good quality) passes this bandpass nearly unchanged $\Rightarrow$ human speech can pass this bandpass nearly without any problem.

Hum noise with $f = 50\,Hz$ lies below the lowest frequency of the bandpass ($\hat{=}\ 100\,Hz$). Therefore hum ~~voice~~ noise is strongly suppressed.



12) $H(z) = \dfrac{1-a}{1-az^{-1}}$

$$H(f) = \frac{1-a}{1-a\,e^{-j2\pi f/r}}$$

$$|H(f_c)| = \frac{|1-a|}{|1-a\,e^{-j2\pi f_c/r}|} = \frac{1}{\sqrt{2}} \cdot \underbrace{H(f=0Hz)}_{=1} = \frac{1}{\sqrt{2}}$$

$|1-a| = 1-a$ for stable filters

$$\Rightarrow \frac{1-a}{\left|1-a\cos\left(2\pi\frac{f_c}{r}\right)+j\cdot a\cdot\sin\left(2\pi\frac{f_c}{r}\right)\right|} = \frac{1}{\sqrt{2}}$$

$$\frac{1-a}{\sqrt{\left(1-a\cos\left(2\pi\frac{f_c}{r}\right)\right)^2 + a^2\sin^2\left(2\pi\frac{f_c}{r}\right)}} = \frac{1}{\sqrt{2}}$$

$$\frac{1-a}{\sqrt{1-2a\cos\left(2\pi\frac{f_c}{r}\right)+a^2}} = \frac{1}{\sqrt{2}}$$

$$\frac{1-2a+a^2}{1-2a\cos\left(2\pi\frac{f_c}{r}\right)+a^2} = \frac{1}{2}$$

$$2-4a+2a^2 = 1-2a\cos\left(2\pi\frac{f_c}{r}\right)+a^2$$

$$\frac{1-4a+a^2}{-2a} = \cos\left(2\pi\frac{f_c}{r}\right)$$

$$\arccos\left(\frac{1-4a+a^2}{-2a}\right) = 2\pi\frac{f_c}{r}$$

$$\Rightarrow f_c = \frac{r}{2\pi}\arccos\left(\frac{1-4a+a^2}{-2a}\right)$$

13)

$$h(n_0) = h(0) \cdot e^{-5}$$

$$\Rightarrow (1-a) \cdot a^{n_0} = (1-a)\, e^{-5}$$

$$\Rightarrow a^{n_0} = e^{-5}$$

$$\Rightarrow n_0 = \log_a e^{-5}$$

$$= \frac{\log_e e^{-5}}{\log_e a}$$

$$= \frac{-5}{\log_e a}$$

15) $H(z) = \dfrac{z^2 - 4z - 7}{z^2 - 9}$

Zeros : $z^2 - 4z - 7 = 0$

$\Leftrightarrow \quad z_{1,2} = -\frac{-4}{2} \pm \sqrt{\left(\frac{-4}{2}\right)^2 + 7} = 2 \pm \sqrt{11}$

poles : $z^2 - 9 = 0$

$(z-3)(z+3) = 0$

$z_{1,2} = \pm 3$